Predicting NFL Team Win Totals using Roster and Activity Data

Team ID: Team 17

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Question and Scope

Question: How accurately can a team's season win total be predicted from its raw roster and activity data using kNN and other predictive modeling?

Variables: Weekly roster turnover, trades, draft capital, strength of schedule, salary cap, dead money

Wins: NFL wins from the schedule data

Scope: All 32 NFL teams in the past 10 years, using teams' reported activity (rosters and team information, combine results, depth charts, etc.)

Data fit and readiness

- ***Fit***: The nflverse tables (schedules, rosters, trades, draft picks, combine) contain the game results and roster data needed to predict next season's win %. The data can address the question because it lets us build turnover and roster features for every team and test whether they predict later wins.
- ***Credibility and Appropriateness***: nflverse is an open source, community maintained GitHub project widely used in NFL analytics, and its data is raw and publicly documented.
- ***Accessibility:*** Loaded directly in R with nflreadr, or downloaded as .csv or .parquet files from the nflverse GitHub. We will be using the .csv files.
- ***Unit of observation***: One row per team season, about 32 teams × ~20+ seasons. Win % is a proportion, and the predictors are counting variables like trades, draft picks and measurements like 40 yard dash in seconds.

Next steps and fallback

Next Steps:

- Pull all raw data
- Define the row and targets:
  - One row per team season
  - Target: next season win %
  - Compute wins from the previous games as the test variable
- Set Baselines: train on past seasons' wins
- Fit models for predictions
- Test Hypothesis: i.e. looking at residuals
- Report Results

Challenges

- Finding the best prediction models for our expected results
- Keeping updated data included with the 2026 season.
- Requires a lot of data (2640 total games across 32 teams and 10 seasons), so we can reduce the scope or number of variables if needed

[Q&A]

AI Usage Disclosure

- AI tool/model: Claude Opus 5
- Used for: Brainstorming ideas and proposal framework and looking generally at our databases' breadth of information
- How AI outputs were evaluated: Weighed against project requirements and criteria, as well as our interest in the ideas